Sensor Ablation Study: Supervised and Unsupervised Wellbeing Models

Ergonomics Monitoring Project

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Abstract

This report analyses two sensor-ablation experiments that quantify how much each sensor group (Posture, Blink, Heart) contributes to wellbeing classification. Both experiments use the same data, the same three sensor groups, the same Leave-One-Subject-Out (LOSO) protocol, and the same three scenarios (*full*, *clean*, *masked*), so their results are directly comparable. The supervised experiment uses a Random Forest classifier; the unsupervised experiment uses Gaussian Mixture clustering whose clusters are mapped to labels only *after* training. We first explain the supervised results, then the unsupervised results, and finally compare the two.

1 Shared experimental setup

Sensor groups (18 features).

- **Posture** (body camera): 9 features — shoulder, head-tilt and forward-head-posture statistics.
- **Blink** (face camera): 7 features — EAR, PERCLOS, blink count, blink frequency, inter-blink intervals.
- **Heart** (wearable): 2 features — `HR` and `RMMSD`.

LOSO validation. Both experiments are validated Leave-One-Subject-Out on the participant column (`video_source`): each fold trains on three participants and tests on the one held-out participant, who is never seen during training. Predictions from all folds are aggregated before computing metrics. LOSO answers the deployment question *"does the model generalise to a brand-new person?"*, which matches the real-time GUI, where incoming windows come from subjects that were not in the training videos.

The three scenarios.

- **full**: train and test with all 18 features. Reference baseline.
- **clean**: train *and* test using only the kept-sensor features. This models the realistic "we simply do not have this sensor" product decision, and is the *trustworthy* ablation scenario.
- **masked**: train with all 18 features, then blank the excluded sensor only at test time so the training-fold imputer fills it with the training median. This is a deployment-mismatch stress test (sensor present in training but missing at runtime), reported for analysis only.

Six configurations are evaluated: three pairs (keep two sensor groups) and three singletons (keep one sensor group).

2 Supervised sensor ablation

Model and tasks. A Random Forest classifier (median imputation + Random Forest) is evaluated on four tasks: *binary* (normal vs. abnormal), *subtype* (sleepy vs. stress), *hierarchical* (binary stage then subtype stage, Random Forest → Random Forest), and *direct 3-class* (normal/sleepy/stress). Because the model is trained with the true labels, it directly learns the decision boundary for each task.

Full-feature baseline. With all 18 features the supervised model is strong on every task (Table **??**): the subtype task is even perfectly separated.

Table 1: Supervised full-feature (18) LOSO reference.

Task	Macro F1	Balanced Acc.
binary	0.952	0.958
subtype	1.000	1.000
hierarchical	0.957	0.957
direct 3-class	0.951	0.951

Clean-scenario results. Table **??** reports the *clean* Macro F1 for every configuration (the reliable scenario). Two clear patterns emerge.

Table 2: Supervised *clean* Macro F1 per configuration and task. Parentheses give the change vs. the full-feature baseline.

Kept sensors	binary	subtype	hierarchical	direct 3-class
Full (18)	0.952	1.000	0.957	0.951
Posture + Blink *(drop Heart)*	0.954 (+0.002)	0.693 (-0.307)	0.763 (-0.195)	0.750 (-0.201)
Posture + Heart *(drop Blink)*	0.957 (+0.005)	0.998 (-0.002)	0.960 (+0.003)	0.959 (+0.008)
Blink + Heart *(drop Posture)*	0.859 (-0.093)	0.981 (-0.019)	0.864 (-0.094)	0.879 (-0.072)
Posture only	0.950 (-0.002)	0.579 (-0.421)	0.697 (-0.261)	0.699 (-0.252)
Blink only	0.862 (-0.091)	0.604 (-0.396)	0.654 (-0.303)	0.661 (-0.291)
Heart only	0.802 (-0.150)	0.988 (-0.012)	0.829 (-0.128)	0.836 (-0.115)

1. **Coarse abnormality (binary) does not need the heart sensor.** Dropping Heart leaves the binary Macro F1 essentially unchanged (0.952 → 0.954). Posture alone already reaches 0.950.

2. **Subtype discrimination relies heavily on the heart sensor.** Removing Heart collapses the subtype task (1.000 → 0.693) and drags down both the hierarchical (−0.195) and the direct 3-class (−0.201) tasks. Conversely, keeping Heart (with Posture) preserves near- perfect subtype performance (0.998). Heart-rate and HRV features are the decisive signal for separating *sleepy* from *stress*.

Masked vs. clean. Across almost all configurations the masked scenario is worse than clean, and in several cases it collapses (e.g. Blink+Heart binary masked = 0.399, the model imputes the missing posture features and predicts a single class). The practical message is that imputing a systematically absent sensor at test time is unreliable; if a sensor will be missing, the model should be retrained without it (clean).

3 Unsupervised sensor ablation

Model and the transductive-to-inductive pipeline. The unsupervised experiment uses Gaussian Mixture clustering (median imputation + standard scaling + Gaussian Mixture). The labels are *not* used during clustering; they are used only afterwards to map each cluster to a wellbeing class by majority vote. To match the GUI deployment — where a model is trained once offline and then applied to real-time windows from unseen subjects — the experiment is run under LOSO: per fold the mixture and the preprocessing are fit on the training participants only, the cluster→label mapping is frozen from the *training* labels, and the frozen model then predicts the held-out participant. For a Gaussian Mixture, fitting and predicting coincide on the training points, so training is effectively a one-shot fit on a fixed set; LOSO is simply the honest estimate of how that frozen model behaves on a new person. Only two tasks are meaningful without supervised subtype labels: *binary* and *direct 3-class*. Because the deployed model uses raw features, no participant z-score representation is included.

Full-feature baseline. With all 18 features (Table **??**) the unsupervised model is, as expected, weaker than the supervised one — it has no labels to shape the decision boundary — and it struggles in particular with the 3-class task.

Table 3: Unsupervised full-feature (18) LOSO reference.

Task	Macro F1	Balanced Acc.
binary	0.877	0.857
direct 3-class	0.700	0.718

Clean-scenario results. Table **??** reports the *clean* Macro F1 for every configuration.

Table 4: Unsupervised *clean* Macro F1 per configuration and task. Parentheses give the change vs. the full-feature baseline.

Kept sensors	binary	direct 3-class
Full (18)	0.877	0.700
Posture + Blink *(drop Heart)*	0.969 (+0.092)	0.671 (-0.029)
Posture + Heart *(drop Blink)*	0.947 (+0.070)	0.687 (-0.013)
Blink + Heart *(drop Posture)*	0.781 (-0.096)	0.647 (-0.052)
Posture only	0.953 (+0.076)	0.589 (-0.111)
Blink only	0.804 (-0.073)	0.550 (-0.150)
Heart only	0.378 (-0.498)	0.483 (-0.216)

1. **Removing the heart sensor *improves* binary clustering.** Posture+Blink reaches 0.969, well above the full-feature 0.877; Posture alone reaches 0.953. The heart features inject participant- specific variance (each person's resting HR/HRV differs), which the label-free clustering partly latches onto, blurring the normal-vs-abnormal split. Dropping them lets the clusters track the wellbeing state instead of the person.

2. **Posture is the backbone; Blink adds little; Heart alone is useless.** Posture alone already yields excellent binary clustering; adding Blink helps marginally (0.953 → 0.969); Heart alone collapses (0.378 binary), because two heart features cluster mostly by participant identity rather than wellbeing.

3. **Removing Heart barely affects the 3-class task (−0.029).** This is *not* because Heart is uninformative for subtypes, but because the unsupervised model cannot separate *sleepy* from *stress* in the first place: the full-feature 3-class score is already only 0.700, and the confusion matrices show heavy sleepy/stress mixing with or without Heart.

Masked vs. clean. As in the supervised case, clean beats masked almost everywhere: for the binary task, dropping Heart cleanly gives 0.969 whereas imputing it at test time (masked) gives only 0.885. If the wearable will be absent, retraining without it is clearly better than imputing it.

4 Comparison of the two experiments

4.1 Where the two methods agree

1. **Binary normal-vs-abnormal detection does not need the heart sensor.** Both methods reach their best (or near-best) binary score after removing Heart (Table **??**). The supervised result is neutral; the unsupervised result is strongly positive. Either way, *posture and blink are sufficient for coarse abnormality detection*, confirming the project's earlier no-heart-sensor finding.

2. **Posture is the single most valuable sensor.** In both experiments, Posture alone gives excellent binary performance (supervised 0.950, unsupervised 0.953), and removing Posture is the most damaging single drop for the binary task (supervised −0.093, unsupervised −0.096).

3. **Blink is the most expendable sensor.** Dropping Blink (keeping Posture+Heart) is essentially free for the binary task in both methods.

4. **Clean is reliable, masked is not.** In both experiments the clean scenario consistently matches or beats masked, and masked degenerates in several configurations. The operational recommendation is identical: if a sensor is going to be missing, retrain without it rather than impute it.

Table 5: Effect of removing the heart sensor (configuration Posture+Blink), full → clean.

Task	Supervised (RF)	Unsupervised (GMM)
binary	0.952 → 0.954 (+0.002)	0.877 → 0.969 (+0.092)
direct 3-class	0.951 → 0.750 (-0.201)	0.700 → 0.671 (-0.029)

4.2 Where the two methods diverge

The clear divergence is the role of the heart sensor for the *subtype / 3-class* distinction:

- In the **supervised** model, Heart is *critical* for separating sleepy from stress: removing it collapses the subtype task (1.000 → 0.693) and the 3-class task (−0.201).
- In the **unsupervised** model, removing Heart costs almost nothing on the 3-class task (−0.029).

This is not a contradiction about the underlying data. The unsupervised model *never* separates sleepy from stress well, even with all sensors (3-class full = 0.700, vs. 0.951 supervised). Because it does not exploit the heart signal in the first place, it cannot lose accuracy when that

signal is removed. The supervised model, having labels, *does* exploit the heart signal, so its removal is keenly felt. In short: heart-rate / HRV features genuinely carry the sleepy-vs-stress information, but only the supervised model is able to use it.

A second, expected divergence is *Heart alone*: the supervised model still extracts a usable binary signal from two heart features (0.802), whereas the unsupervised model collapses (0.378), because without labels two heart features cluster by participant identity rather than wellbeing state.

4.3 Interpretation note: different scientific questions

The two experiments answer related but distinct questions. Supervised LOSO asks *"can a trained classifier predict an unseen participant's state from this sensor subset?"* Unsupervised LOSO asks *"do the natural clusters in this sensor subset align with wellbeing states for an unseen participant?"* The strong agreement on the binary, posture, and blink findings is therefore meaningful: it holds whether or not the model is given labels. The divergence on subtypes should be read as a statement about *what each method can use*, not as disagreement about the sensors themselves.

5 Conclusion

Both the supervised and the unsupervised sensor-ablation experiments, run under identical LOSO conditions, converge on the deployment-relevant conclusions:

- **Coarse abnormality detection (binary) is robust without the wearable.** Posture and blink are sufficient; in the unsupervised model the heart sensor is even mildly harmful, as it carries participant-specific variance.
- **Posture is the key sensor and blink the most expendable.**
- **Heart-rate / HRV features carry the sleepy-vs-stress signal,** but only the supervised model can exploit it; the unsupervised model does not separate the subtypes regardless.
- **Clean > masked.** A sensor that will be absent at runtime should be removed from training, not imputed.

For a product that only needs normal-vs-abnormal monitoring, the heart wearable can be dropped with no loss (supervised) or even a gain (unsupervised). For sleepy-vs-stress discrimination, the heart wearable and a supervised model are both required.